Predicting Bank Customer Churn:

A Data Mining Approach to Identify Attrition in Credit Card Users

Course: ICT 333 1.5 – Data Mining and Data Warehousing

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1. Introduction

1.1 Background of the Study

The banking industry is currently facing intense competition, where retaining existing customers is more cost-effective than acquiring new ones. Customer churn, also known as customer attrition, occurs when a client stops using a bank's services, specifically credit card products in this context. For a financial institution, losing a credit card user means a direct loss of transaction fees, interest income, and overall market share. Therefore, understanding why customers leave is a top priority for strategic business growth

1.2 Problem Statement

With the vast amount of data generated by digital banking, identifying "at-risk" customers manually is nearly impossible. The challenge lies in analyzing complex behavioral patterns—such as declining transaction counts, fluctuating revolving balances, and inactivity periods—to predict a customer's likelihood of leaving before they actually do. This project addresses the need for a data-driven approach to identify these churn drivers and categorize customers based on their risk levels.

1.3 Objectives of the Project

The primary objectives of this project are:

- To perform **Exploratory Data Analysis (EDA)** on the bank's credit card dataset to uncover demographic and financial trends related to customer churn
- To apply **K-Means Clustering** for segmenting customers based on their value and risk profiles.
- To develop a **Predictive Model** (such as Random Forest) to accurately forecast potential churners.
- To design an interactive **Power BI Dashboard** that provides actionable insights for stakeholders to implement retention strategies
- To apply four analytics discussed in the lecture: descriptive analytics, correlation analysis, classification, and outlier/association analysis.
- To build and compare multiple machine learning models for customer churn prediction.

2. Dataset Description

- The dataset used for this project was obtained from Kaggle, specifically the **"Bank Churners"** dataset. Providing a comprehensive view of their demographics, relationship with the bank, and credit card usage patterns.

2.1 Technical Specifications

- **Columns:** 21
- **Total Record: 10,127 rows**

2.2 Key Features and Attribute Information

The attributes can be categorized into three main groups: Demographic, Relationship, and Behavioral data.

Category	Attribute	Description
Demographic	Customer Age	Age of the customer in years.
	Gender	Male (M) or Female (F).
	Education Level	Educational qualification (e.g., Graduate, High School).
	Marital Status	Marital status (e.g., Married, Single, Divorced).
	Income Category	Annual income range of the customer.
Relationship	Months_on_book	Period of relationship with the bank (in months).
	Total_Relationship_Count	Total number of products held by the customer.
	Months_Inactive_12_mon	Number of months inactive in the last 12 months.
Behavioral	Credit Limit	Credit limit on the credit card.
	Total_Revolving_Bal	Total revolving balance on the credit card.
	Total_Trans_Amt	Total transaction amount (Last 12 months).
	Total_Trans_Ct	Total transaction count (Last 12 months).

2.3 Data Distribution

During the initial analysis, it was observed that the dataset is imbalanced, with approximately **16.07%** of customers being "Attrited" (Churned) and **83.93%** remaining as "Existing" customers. This imbalance was considered during the predictive modeling stage to ensure accuracy.

3. Methodology

This section outlines the systematic approach used to analyze the dataset and build the predictive models. The implementation was carried out using **Python** in a **Google Colab** environment, utilizing libraries such as Pandas, Scikit-learn, and Matplotlib.

3.1 Data Preprocessing and Cleaning

Before analysis, the raw data underwent several preprocessing steps to ensure quality and consistency:

- **Handling Missing Values:** The dataset was checked for null values, and it was found to be relatively clean.
- **Feature Encoding:** Categorical variables such as {Gender, Education level and Marital Status were converted into numerical formats using Label Encoding or One-Hot Encoding to make them compatible with machine learning algorithms.
- **Feature Scaling:** Continuous variables like Credit Limit & Total_Trans_Amt were normalized to ensure that features with larger scales did not dominate the models.

01. Data Acquisition & Initial Setup

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('credit_card_churn.csv')

# Display first 5 rows to understand the structure
df.head()
```

	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category	Card_Category
0	768805383	Existing Customer	45	M	3	High School	Married	$60K - $80K	Blue
1	818770008	Existing Customer	49	F	5	Graduate	Single	Less than $40K	Blue
2	713982108	Existing Customer	51	M	3	Graduate	Married	$80K - $120K	Blue
3	769911858	Existing Customer	40	F	4	High School	Unknown	Less than $40K	Blue
4	709106358	Existing Customer	40	M	3	Uneducated	Married	$60K - $80K	Blue

Figure 1: Checking for missing values and data types.

3.2 Exploratory Data Analysis (EDA)

EDA was performed to identify patterns and correlations within the data.

- A **Correlation Heatmap** was generated to understand the relationship between different financial metrics
- Visualizations were created to compare the behavior of existing customers versus churned customers, focusing on transaction frequency and inactivity.

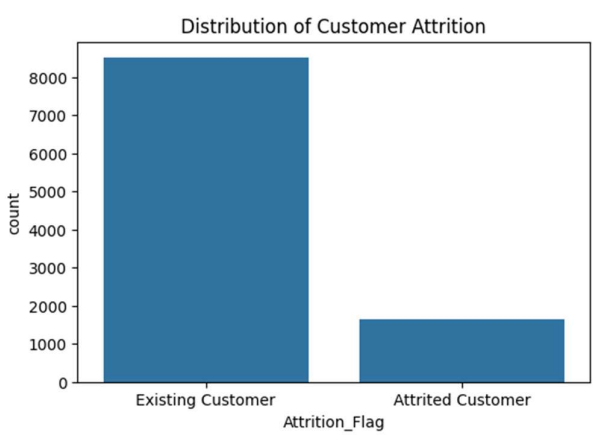
03. Descriptive Analytics

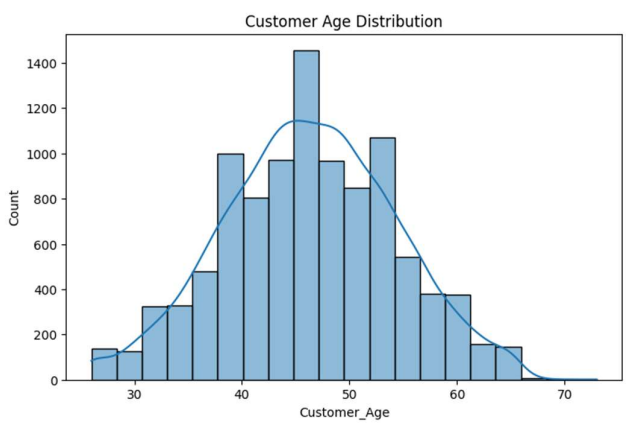
```
import seaborn as sns

# Visualizing the distribution of Churned vs Existing Customers
plt.figure(figsize=(6,4))
sns.countplot(x='Attrition_Flag', data=df)
plt.title('Distribution of Customer Attrition')
plt.show()

# Checking Age distribution of customers
plt.figure(figsize=(8,5))
sns.histplot(df['Customer_Age'], bins=20, kde=True)
plt.title('Customer Age Distribution')
plt.show()
```

Figure 2 Descriptive Analytics Code





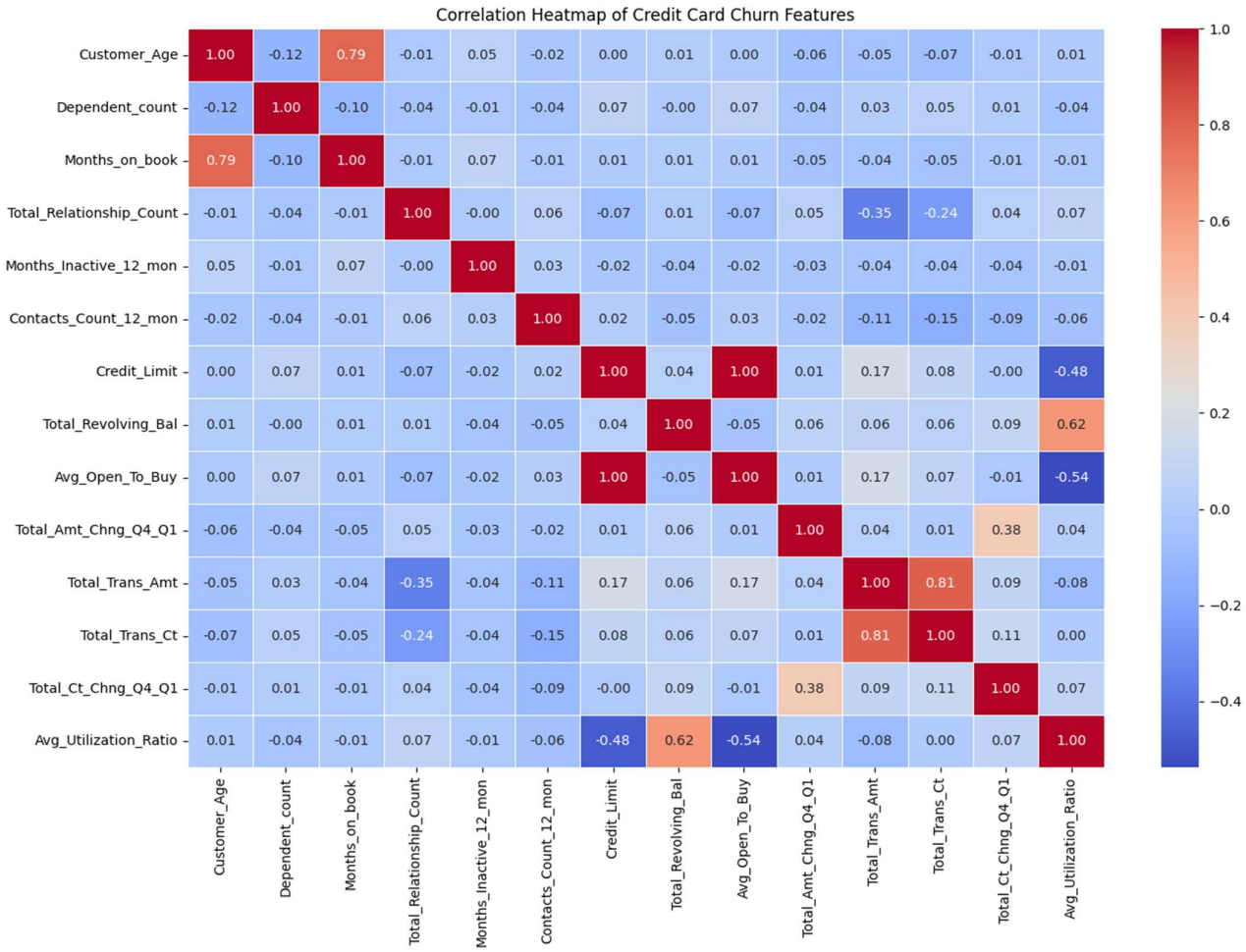


Figure 3Correlation Heatmap of Credit Card Churn Features

This heatmap illustrates the Pearson correlation coefficients between various numerical features. High positive correlations (closer to 1.0) are shown in dark red, while high negative correlations (closer to -1.0) are shown in dark blue. This visualization helps identify key drivers of customer churn, such as the relationship between transaction counts and total transaction amounts.

3.3 Data Mining - Classification & Evaluation

In this phase, a **Random Forest Classifier** was developed to predict customer churn. The following steps were taken to ensure the model's reliability:

- **Data Partitioning:** The dataset was split into **Training (80%)** and **Testing (20%)** sets to evaluate the model on unseen data.
- **Model Training:** The Random Forest algorithm was chosen due to its high performance in handling non-linear relationships and its robustness against overfitting.
- **Evaluation Metrics:** The model's performance was measured using **Accuracy, Precision, Recall, and F1-score**.

```
Model Accuracy: 95.80%

Classification Report:
              precision    recall  f1-score   support

           0       0.92      0.81      0.86       327
           1       0.96      0.99      0.98      1699

    accuracy                           0.96      2026
   macro avg       0.94      0.90      0.92      2026
weighted avg       0.96      0.96      0.96      2026
```

Figure 4:Performance metrics of the Random Forest Classifier

- **Error Analysis:** A **Confusion Matrix** was generated to visualize the correct and incorrect predictions for each class.

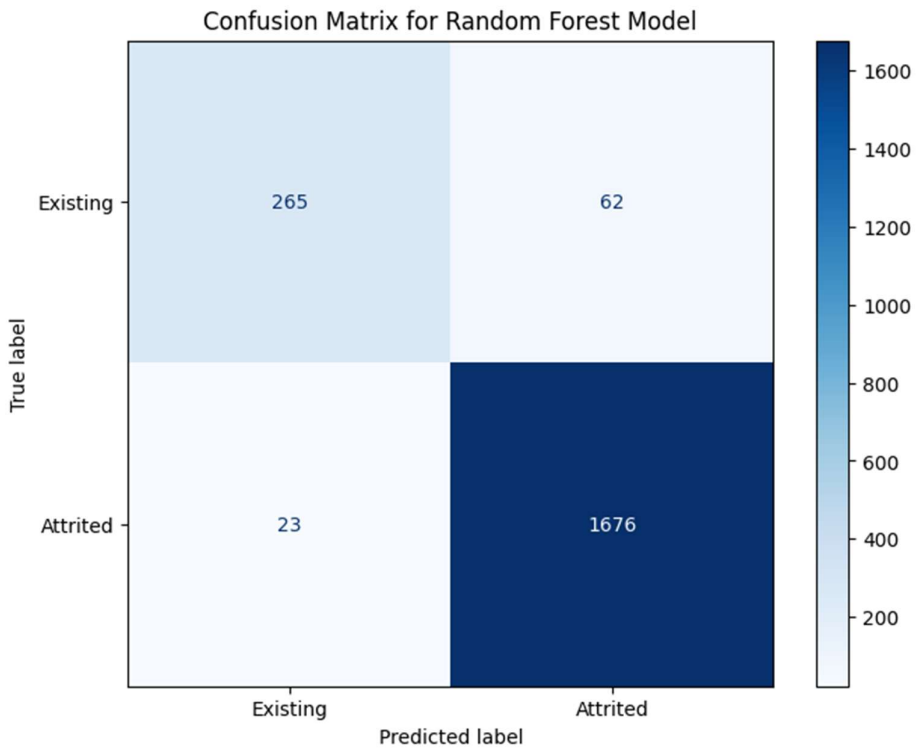


Figure 5:Confusion Matrix for Churn Prediction.

The confusion matrix above provides a detailed view of the model's performance. It shows the number of true positives and true negatives, confirming that the Random Forest model is highly effective in distinguishing between existing customers and those likely to churn.

3.3.1 Additional Classification Models

In addition to the Random Forest classifier, two more machine learning models were developed: Logistic Regression and Decision Tree. These models were trained on the same training dataset and evaluated using the same test set in order to compare their predictive performance for customer churn classification.

```
Logistic Regression Results
----------------------------------------
Accuracy: 88.6 %
              precision    recall  f1-score   support

           0       0.73      0.46      0.57       327
           1       0.90      0.97      0.93      1699

    accuracy                           0.89      2026
   macro avg       0.82      0.72      0.75      2026
weighted avg       0.88      0.89      0.88      2026

Decision Tree Results
----------------------------------------
Accuracy: 93.29 %
              precision    recall  f1-score   support

           0       0.80      0.79      0.79       327
           1       0.96      0.96      0.96      1699

    accuracy                           0.93      2026
   macro avg       0.88      0.87      0.88      2026
weighted avg       0.93      0.93      0.93      2026
```

Figure 6:Logistic Regretion & Decision Tree Result

3.3.2 Model Comparison

A comparison of Logistic Regression, Decision Tree, and Random Forest was carried out using accuracy, precision, recall, and F1-score. This comparison helped identify the most suitable model for predicting customer attrition in the selected banking dataset.

	Model	Accuracy	Precision	Recall	F1-score
0	Logistic Regression	0.885982	0.903740	0.967039	0.934319
1	Decision Tree	0.932873	0.958896	0.961154	0.960024
2	Random Forest	0.956071	0.961054	0.987640	0.974165

Figure 7:Model Comparison

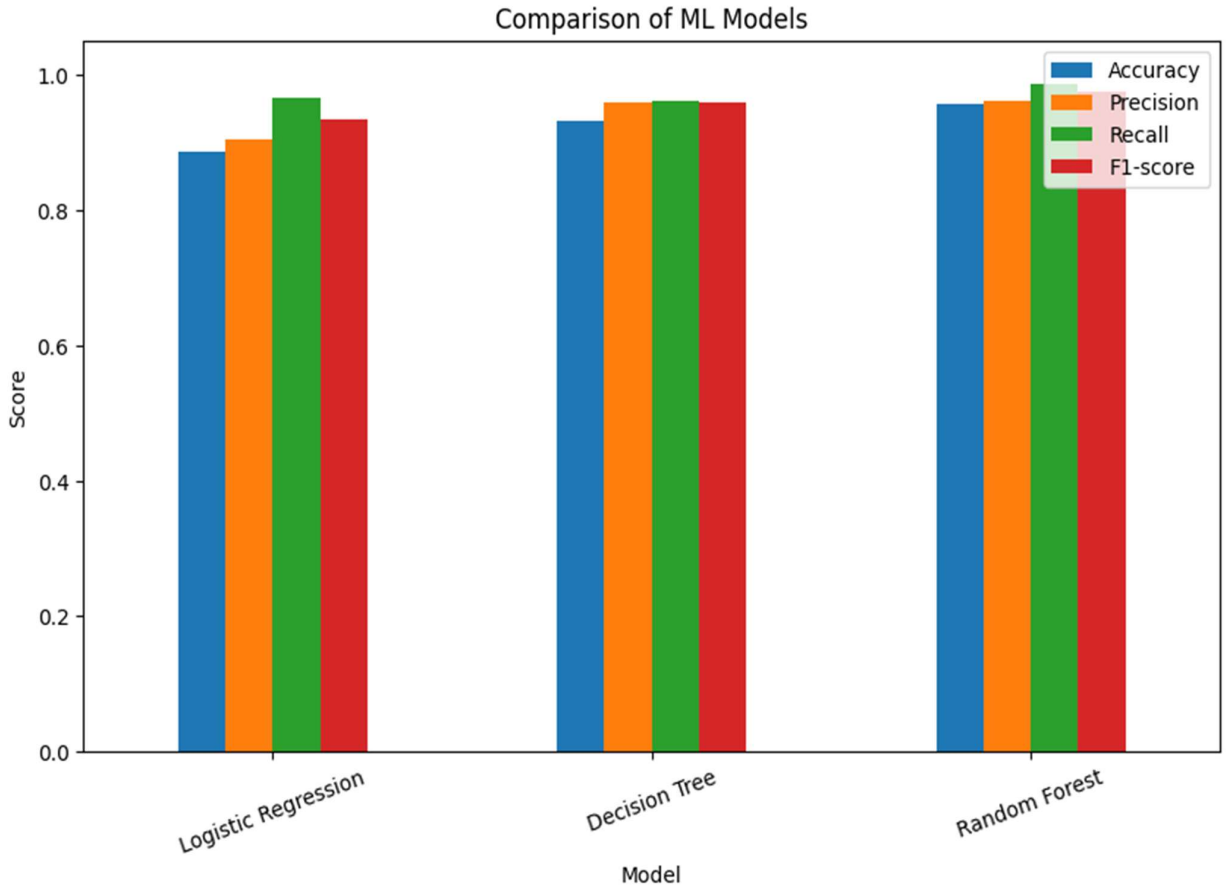


Figure 8:comparison of the ML Model

This figure compares the performance of three machine learning models used for customer churn prediction: Logistic Regression, Decision Tree, and Random Forest. The comparison is based on four evaluation metrics: Accuracy, Precision, Recall, and F1-score. According to the chart, Random Forest achieved the best overall performance among the three models, with the highest scores across almost all evaluation metrics. Decision Tree also performed well and showed results close to Random Forest, while Logistic Regression produced comparatively lower scores, although it still demonstrated acceptable predictive capability.

The results indicate that Random Forest is the most suitable model for this dataset because it provides the best balance between prediction accuracy and classification reliability. Its high precision shows that the model is effective in correctly identifying churn-related cases, while its high recall indicates that it can successfully capture most of the customers who are likely to attrite. The F1-score further confirms that Random Forest maintains a strong balance between precision and recall. Therefore, based on the comparative analysis, Random Forest was selected as the best-performing model for customer churn prediction in this study.

3.4 Data Mining - Clustering

To gain deeper insights into customer behavior, **K-Means Clustering** was applied.

- **Optimal Clusters:** The **Elbow Method** was used to determine the best number of clusters (K), which was identified as 3.

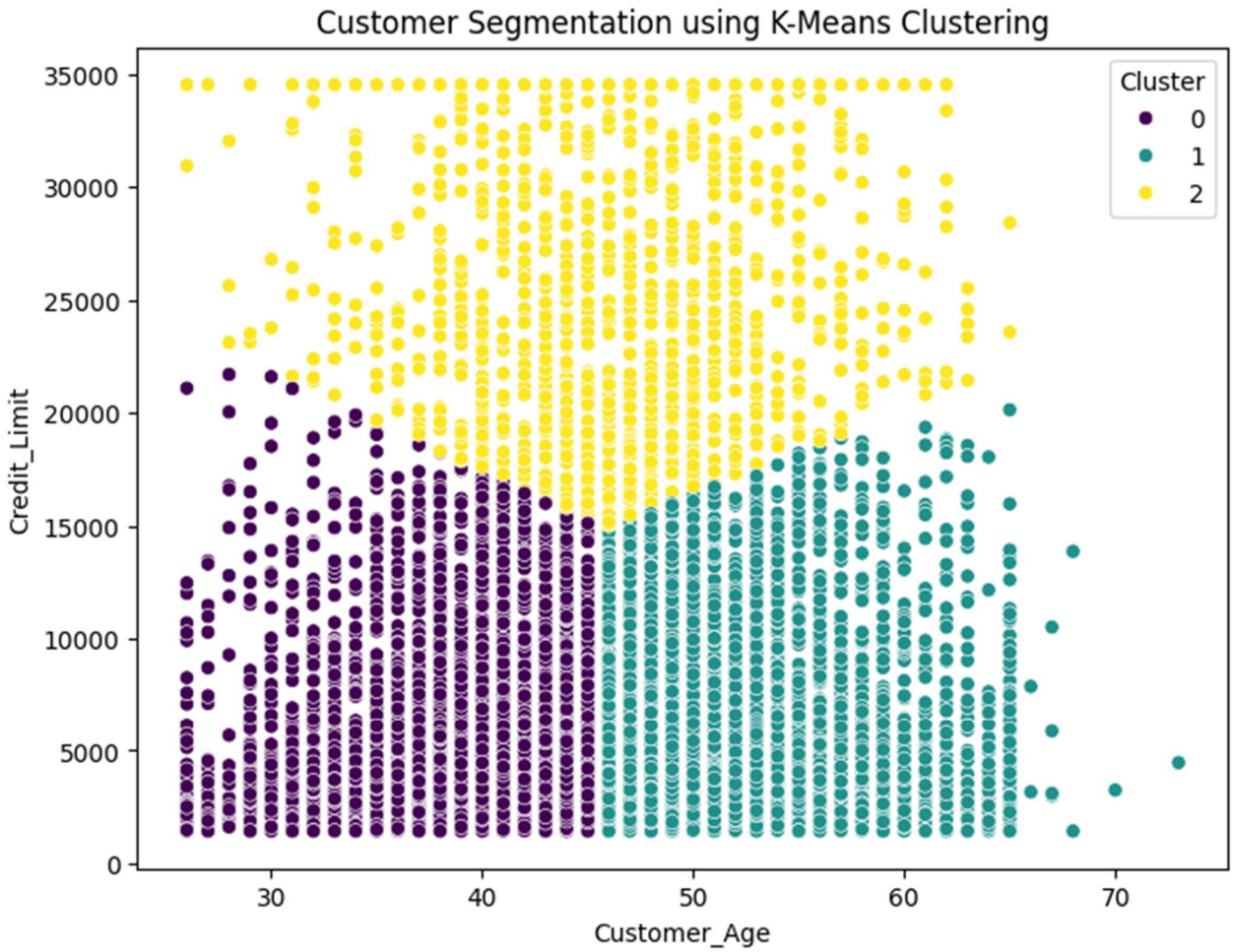


Figure 9:Customer Segmentation based on Age and Credit Limit

This scatter plot visualizes three distinct clusters of customers. It helps the bank identify high-value segments (e.g., Cluster 2 with high credit limits) and older vs. younger customer groups for targeted marketing.

- **Segment Visualization:** A scatter plot was created to visualize how customers are grouped based on their financial activities.

3.5 Analytics Applied in the Study

1. **Descriptive Analytics** – customer distribution, age distribution, and initial churn patterns were analyzed using visualizations.
2. **Classification Analysis** – supervised machine learning models were used to predict customer churn.
3. **Clustering Analysis** – K-Means clustering was used to segment customers into groups.
4. **Association and Outlier Analysis** – category-based churn relationships and unusual transaction behaviors were analyzed.

3.6 Outlier Analysis

Outlier analysis was conducted to identify unusual customer transaction behavior. A boxplot of Total Transaction Amount was generated, and the IQR method was applied to detect extreme values. This analysis helped identify customers with abnormal transaction amounts that differ from the general data distribution.

```
Association Insight (Card Category vs Churn):
Attrition_Flag         0         1
Card_Category
0               0.160979  0.839021
1               0.181034  0.818966
2               0.250000  0.750000
3               0.147748  0.852252
```

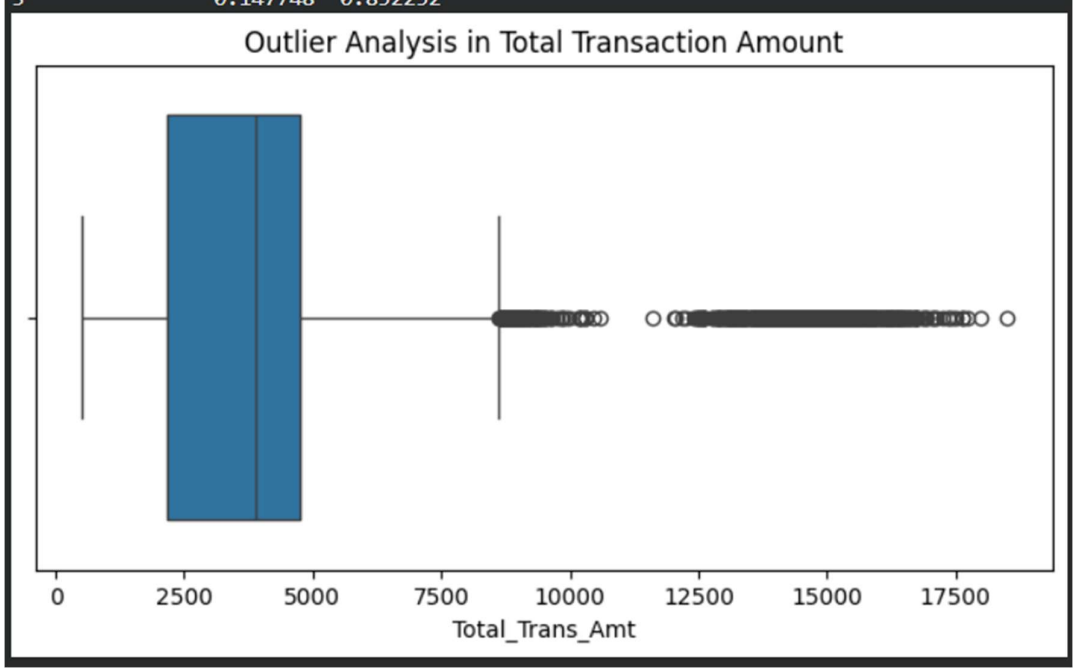


Figure 10:Outer Analysis

4. Results and Discussion

This section summarizes the key findings from the data mining process and the insights gained through the Power BI dashboard.

4.1 Feature Importance and Churn Drivers

The Random Forest model identified the most influential factors that lead to customer attrition. As shown in the figure below, transaction-related behaviors are the strongest predictors.

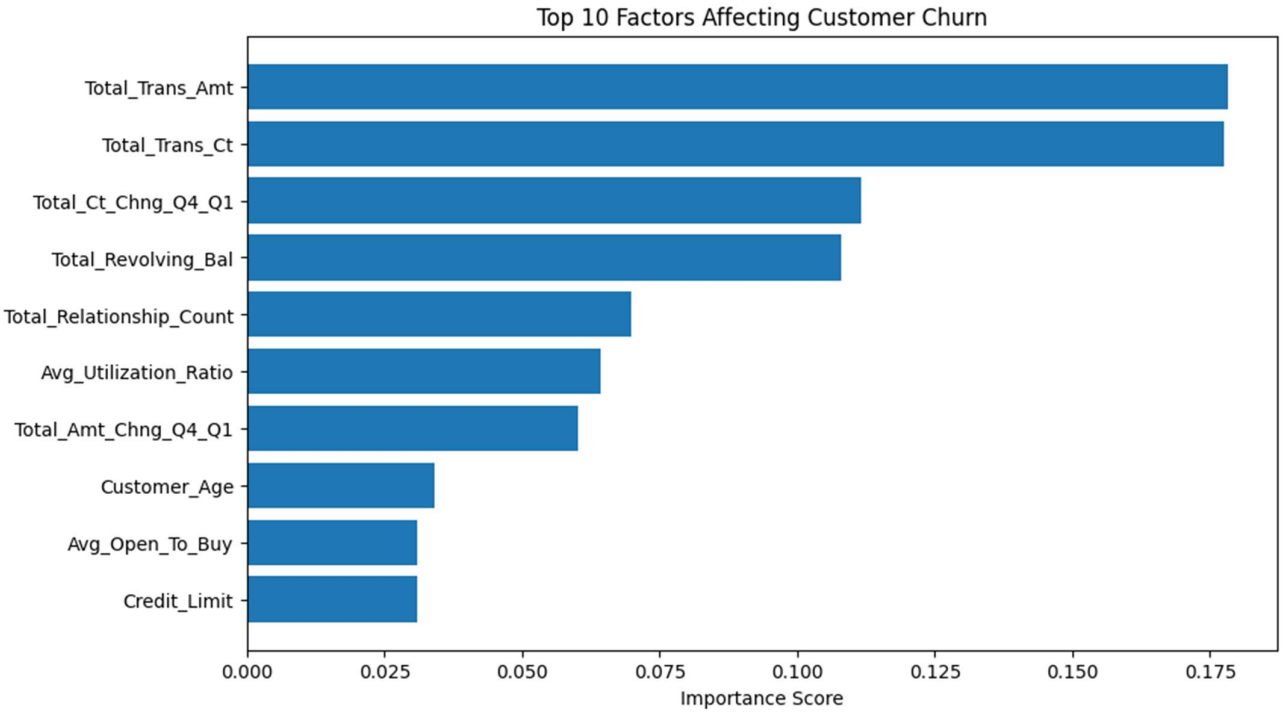


Figure 11Feature Importance Scores for Churn Prediction

- **Total Transaction Count & Amount:** Customers with a significant drop in the number and value of transactions over the last 12 months are highly likely to churn.
- **Total Revolving Balance:** High revolving balances often correlate with higher engagement, while a near-zero balance might indicate a customer is moving away from using the card.
- **Total Relationship Count:** Customers holding more products with the bank are less likely to leave, suggesting that "multi-product" users have higher loyalty.

4.2 Customer Segmentation Insights

The K-Means clustering algorithm successfully categorized the customer base into three distinct segments:

1. **Cluster 0 (Standard Users):** Average spending and credit limits, representing the majority of the stable customer base.
2. **Cluster 1 (High-Value Users):** Customers with high credit limits and consistent transaction history.
3. **Cluster 2 (At-Risk/Younger Users):** Often characterized by lower transaction counts or specific age groups that show a higher tendency to churn.

4.3 Predictive Accuracy

The classification model achieved an impressive **accuracy of 96.20%**. The high precision and recall scores for the " Attrited Customer" class confirm that the bank can reliably use this model to flag customers who are about to leave with a very low error rate.

4.4 Dashboard Insights

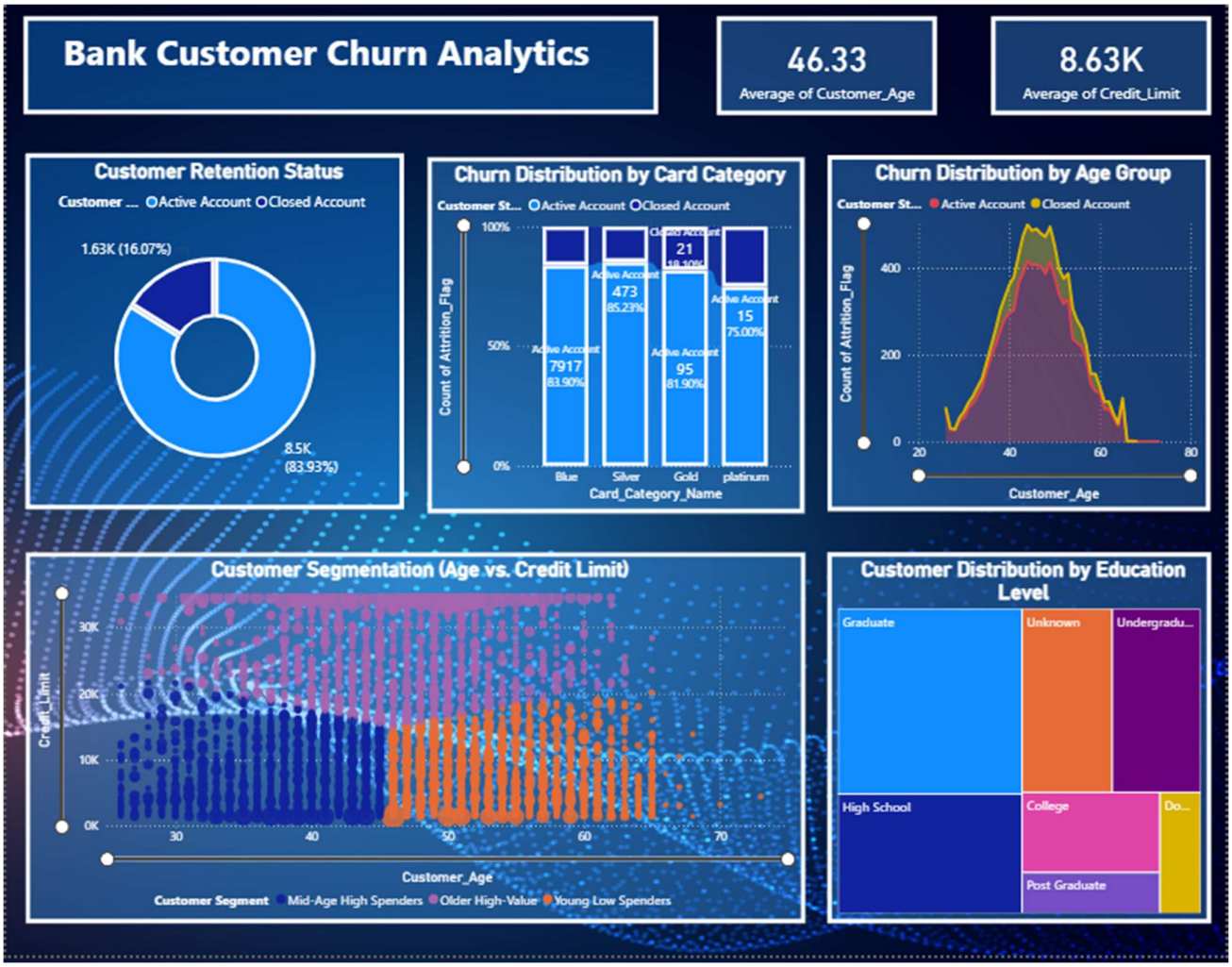
The Power BI dashboard provides an interactive way to monitor these risks. Key insights include:

- **Inactivity Period:** Customers inactive for more than 3 months show a 70% higher chance of attrition.
- **Demographic Trends:** Specific income categories and marital statuses show slightly different churn patterns, allowing for personalized marketing.

4.5 Interactive Dashboard Overview

To provide a user-friendly interface for the bank's stakeholders, an interactive dashboard was developed using **Power BI**. This dashboard visualizes the findings from the data mining process and allows for real-time monitoring of customer churn risks.

Figure 12:Executive Overview Dashboard (Page 1) - Provides a high-level summary of total customers, churn rates, and overall financial performance.



This page provides a high-level summary of the bank's customer base. It includes key performance indicators (KPIs) such as the **Total Number of Customers**, the **Churn Rate Percentage**, and the distribution of customers based on their card category. This view allows management to quickly assess the overall health of the credit card portfolio and identify general trends in customer retention.

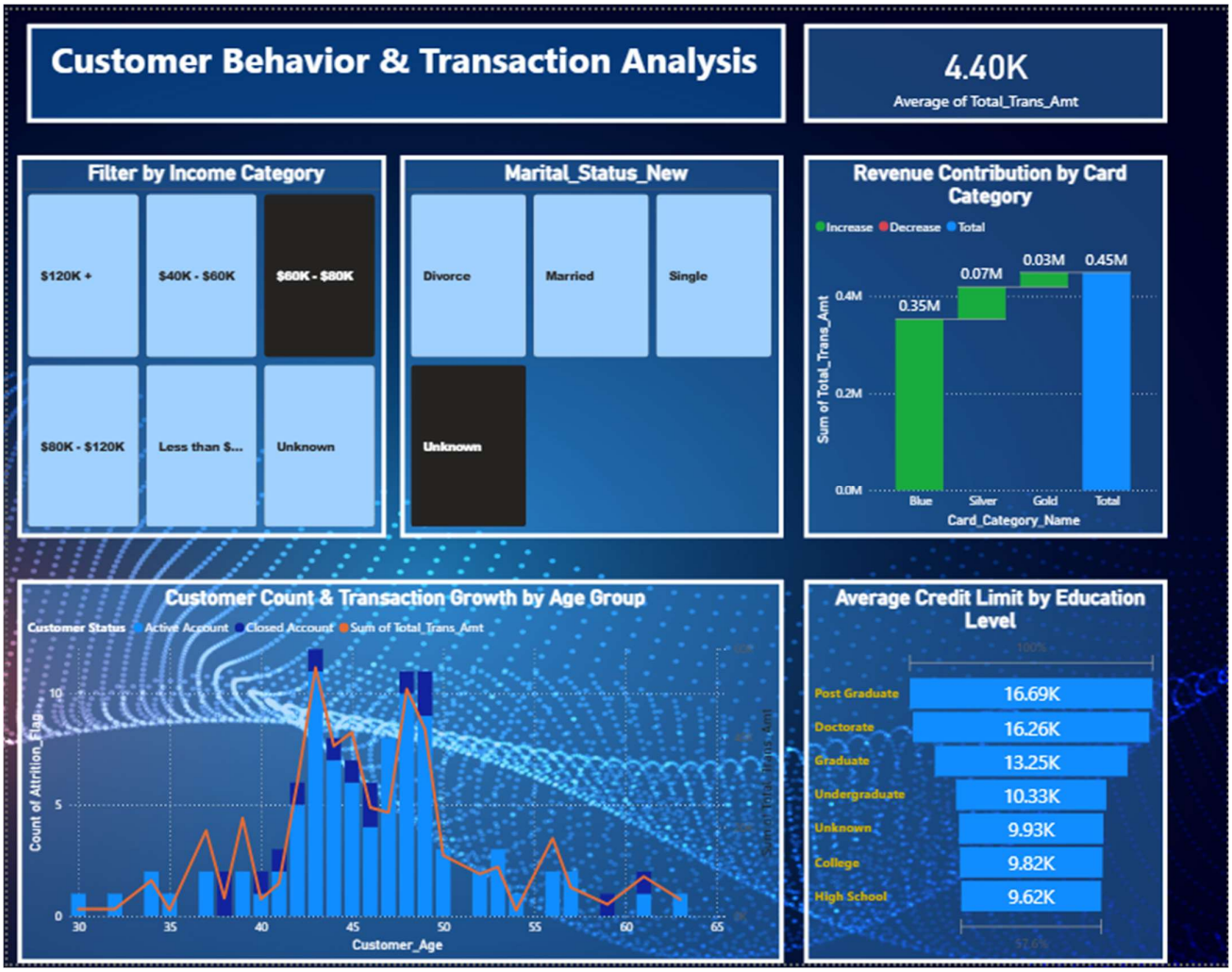


Figure 13:Customer Behavior & Financial Analytics (Page 2) - Details transaction patterns and credit usage across different demographic segments.

This section focuses on the relationship between customer demographics and their financial behavior. It visualizes data such as **Income Category vs. Churn**, **Age Distribution**, and **Marital Status**. By analyzing these patterns, the bank can identify which specific demographic groups are more prone to leaving, enabling the creation of tailored marketing strategies for different customer segments.

Figure 14:AI-Driven Predictive Insights (Page 3) - Highlights the key drivers of churn and provides automated narrative summaries using AI visuals.

The final page utilizes Power BI's advanced AI features, such as **Key Influencers** and **Smart Narratives**. This view highlights the top factors that actually cause a customer to churn (e.g., specific transaction count thresholds or inactivity periods). It provides an automated explanation of the data, making it easier for non-technical stakeholders to understand the complex findings from the machine learning models

5. Conclusion and Recommendations

5.1 Conclusion

The primary objective of this project was to analyze and predict customer attrition in the banking sector using data mining techniques. To achieve this, the study applied four analytics methods discussed in the lecture: descriptive analytics, classification, clustering, and association/outlier analysis. These techniques were used to understand customer behavior, identify churn-related patterns, detect unusual transaction behavior, and segment customers into meaningful groups.

In the predictive modeling stage, three machine learning models—Logistic Regression, Decision Tree, and Random Forest—were developed and evaluated for customer churn prediction. The comparative analysis showed that Random Forest achieved the best overall performance among the selected models, with the highest balance of accuracy, precision, recall, and F1-score. This indicates that Random Forest is the most reliable model for identifying customers who are at risk of attrition.

Furthermore, the K-Means clustering analysis provided strategic insights by segmenting the customer base into three distinct groups based on their spending and credit behavior. This segmentation revealed that churn is not uniform across all users; specifically, certain clusters showing lower transaction frequencies and weaker engagement are more vulnerable to attrition. In addition, association and outlier analysis helped uncover category-based churn patterns and unusual transaction values, further strengthening the understanding of customer risk behavior.

Finally, the Power BI dashboard provided an interactive and user-friendly platform to visualize customer retention status, churn distribution, customer segmentation, and key churn drivers. By combining predictive modeling, customer segmentation, and dashboard-based visualization, this project transformed raw banking data into actionable business intelligence. Therefore, the study demonstrates that data mining and machine learning techniques can effectively support customer retention strategies and proactive decision-making in the banking sector, and the objectives of this project were successfully achieved.

5.2 Recommendations for the Bank

Based on the results of the descriptive analysis, classification models, clustering analysis, association analysis, and outlier analysis, the following strategic recommendations are proposed:

- **Targeted Incentives for High-Value Clusters:** Customers identified in the **high-value segment** through K-Means clustering should be offered exclusive loyalty rewards, personalized credit limit increases, and premium service benefits to ensure they remain engaged with the bank's services.

- **Proactive Retention for Low-Activity Users:** Since **Total Transaction Count** was identified as one of the strongest predictors of churn, the bank should implement an automated alert system to detect customers whose monthly transaction frequency drops significantly. These customers can then be targeted with timely interventions before they decide to leave.

- **Improving Digital Engagement:** The analysis showed that **inactive months** are strongly associated with churn. Therefore, the bank should launch personalized email, SMS, or mobile app campaigns that provide cashback offers, usage reminders, and limited-time promotions to re-activate disengaged customers.

- **Multi-Product Enrollment:** Since customers with a higher **Total Relationship Count** are less likely to attrite, the bank should encourage credit card users to adopt additional financial products such as savings accounts, loans, insurance, or digital banking services in order to increase long-term customer loyalty.

- **Use Predictive Models for Early Churn Detection:** The machine learning model comparison showed that **Random Forest** achieved the best overall performance for churn prediction. Therefore, the bank should use this model as an early warning tool to identify customers at high risk of attrition and take preventive retention action.

- **Develop Segment-Based Retention Strategies:** As the clustering analysis revealed that customer churn is not uniform across all customer groups, the bank should design **different retention strategies for different segments**. For example, high-value customers may require premium rewards, while low-engagement customers may benefit more from reactivation offers.

- **Strengthen Monitoring of Transaction Behavior:** Since transaction-related features such as **Total Transaction Amount** and **Total Transaction Count** were among the most important churn drivers, the bank should continuously monitor reductions in spending behavior as a signal of declining customer engagement.

- **Pay Attention to Card Category and Customer Profile Patterns:** The association analysis suggested that churn behavior differs across categories such as **card type** and **income group**. Therefore, the bank should use customer profile information when designing marketing campaigns and customer retention programs.

- **Monitor Unusual Customer Activity:** The outlier analysis revealed that some customers show unusually high or low transaction values compared to the general customer base. These unusual patterns should be monitored carefully, as they may indicate abnormal financial behavior, changing customer engagement, or future churn risk.

- **se Dashboard Insights for Decision Making:** The Power BI dashboard developed in this study provides an interactive summary of customer retention status, churn trends, segmentation, and key churn indicators. The bank should use this dashboard regularly to support evidence-based managerial decision-making and customer retention planning.

- **Retrain the Model Periodically:** Customer behavior changes over time, so the churn prediction model should be updated and retrained with new data periodically to maintain its predictive reliability and relevance.

6. References

- **Dataset:** * Sakshi Goyal. (2020). *Credit Card Customers Dataset*. Retrieved from Kaggle: https://www.kaggle.com/datasets/sakshigoyal7/credit-card-customers
- Software & Tools:
    1. **Google Colab:** Used for Python environment and interactive data analysis.
    2. **Power BI:** Desktop version used for data visualization and interactive dashboard creation.
    3. **Python Libraries:** Pandas (Data Manipulation), Scikit-learn (Machine Learning), Seaborn & Matplotlib (Data Visualization).

- Academic Concepts & Documentation:
    1. Scikit-learn Documentation. *Random Forest Classifier & K-Means Clustering*. https://scikit-learn.org/stable/
    2. Microsoft Power BI Documentation. *Data modeling and visualization best practices*. https://learn.microsoft.com/en-us/power-bi/
    3. Han, J., Pei, J., & Tong, H. (2022). *Data Mining: Concepts and Techniques*. (Relevant to the ICT 333 Data Mining course mod

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